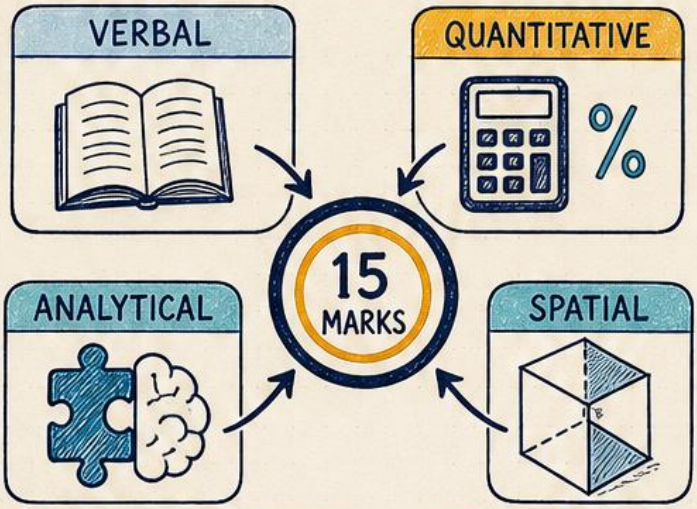
@ codedsoul_05

GATE 2027 • GA ★

GATE 2027 GENERAL APTITUDE

HANDWRITTEN HANDBOOK

Verbal • Quantitative • Analytical • Spatial — 15 Marks That Move Rank



> "Accuracy first.
> Speed second.
> Rank follows."

Grammar • Arithmetic • Logic • DI • Spatial • Strategy

1 WHAT'S INSIDE

1. Verbal Ability
2. Quantitative Ability
3. Analytical Reasoning
4. Spatial Ability
5. Data Interpretation
6. Exam Strategy & Approach
7. PYQs & Practice Sets
8. Shortcuts & Time-savers

2 GATE GA SNAPSHOT

- 15 Marks
- 10 Questions
- 5 × 1 mark + 5 × 2 marks
- Common to all papers

Marking Scheme (MCQ):

- 1-mark MCQ : − 1/3 mark
- 2-mark MCQ : − 2/3 mark

- No Negative Marking for MSQ & NAT Questions

3 TOPICS COVERED

- ☑ Grammar & Usage
- ☑ Vocabulary
- ☑ Comprehension
- ☑ Arithmetic
- ☑ Algebra & Number System
- ☑ Modern Math
- ☑ Logical Reasoning
- ☑ Data Interpretation
- ☑ Data Sufficiency
- ☑ Spatial Ability (2D & 3D)
- ☑ Strategy & Time Management

WHO FOR?

GATE 2027 ASPIRANTS — ALL BRANCHES

> "GA is the highest ROI section per hour of prep."

★ *Sharpen Language. Speed Numbers. See Patterns.* ★

01 — EXAM PATTERN & SYLLABUS

Same GA paper for every GATE discipline — treat it as a free 15 marks.

PATTERN

- Total GA = 15 marks
- 10 questions: 5 × 1 mark + 5 × 2 marks
- Types: MCQ, MSQ, NAT (as per paper mix)
- MCQ negative marking: −1/3 (1-mark), −2/3 (2-mark)
- MSQ & NAT: no negative marking
- Rest of GATE = 85 marks technical

MARKS STRATEGY



Target 12–15/15.

Even 10/15 is strong.

Do NOT leave GA for last week only.

OFFICIAL SYLLABUS (4 pillars)

1 VERBAL APTITUDE

grammar, vocabulary, RC, narrative sequencing

2 QUANTITATIVE APTITUDE

numbers, arithmetic, algebra basics, geometry/mensuration, stats/probability, DI

3 ANALYTICAL APTITUDE

deduction/induction, analogy, matching, logic puzzles

4 SPATIAL APTITUDE

shape transformations, paper folding, 2D/3D assembly

In the exam, attempt GA early while fresh — but skip sticky RC/spatial and return.

Know the syllabus. Respect negative marks. Bank easy GA. ★

02 — VERBAL: GRAMMAR ESSENTIALS

GATE grammar is pattern recognition — subject-verb, modifiers, tenses, articles, prepositions.

① Subject-Verb agreement

- each/everyone → singular
- either... or → nearer subject

Ex: Each of the players is ready.

Either you or I am responsible.

② Tenses & sequence (if conditional types briefly)

- If + present → future
- If + past → would + V
- If + past perfect → would have + V.

Ex: If it rains, we will cancel.

If he studied, he would pass.

③ Articles a/an/the — unique vs generic

- a/an → non-specific
- the → specific/unique
- Ø (no article) → generic

Ex: I saw a cat.

The sun rises in east.

Cats are mammals.

④ Prepositions

- interested in
- accuse of
- prefer to
- between / among

Ex: She is interested in research.

He was accused of theft.

I prefer tea to coffee.

Between us, keep it secret.

Among the students, he is the tallest.

⑤ Parallelism in lists

- Keep items in same form (noun / verb / adjective).

Ex: He likes reading, swimming and to cook. ✗

He likes reading, swimming and cooking. ✓

⑥ Modifier placement (dangling modifiers)

- Place modifier next to the word it modifies.

Ex: Walking in the park, the flowers looked beautiful. ✗

Walking in the park, I saw beautiful flowers. ✓

⑦ Active / Passive quick transform

- Object of active → subject of passive.

Ex: Active: They built the bridge.

Passive: The bridge was built by them.

⑧ Conjunctions

- although (contrast) vs because (reason)
- since (point of time) vs for (duration)

Ex: Although it rained, we went out.

I stayed because I was tired.

I have lived here since 2010.

I have lived here for five years.

⚠ **COMMON TRAP (AGREEMENT):**

One of the students ARE → IS. The number of X IS; A number of X ARE.

MINI PRACTICE: (Fill in the blank / Error spotting)

1) Each of the engineers _____ (is/are) working on the project.
2) If she _____ (goes/go) now, she will catch the train.
3) He is _____ (an/the/—) honest man.
4) The data was discussed _____ (in/at/on) the meeting.
5) She enjoys music, painting and _____ (write/writing/to write).
6) _____ the heavy rain, the match continued. (No Error / Error)
7) The award was given _____ (to him/by him) by the chief guest.
8) I have known him _____ (since/for) three years.

ANSWERS

1) is
2) goes
3) an
4) in
5) writing
6) Error (Dangling)
7) to him
8) for

★ Agreement, parallelism, prepositions — highest hit rate.

03 — VERBAL: VOCAB, RC & SEQUENCING

You rarely need rare words — you need roots, context, and elimination.

① VOCAB STRATEGIES

- **Roots:** bene, mal, dict, scrib/script, port, spect, ject, cred
- **Prefixes:** un/in/im/il/ir, re, pre, anti, inter, over/under
- **Contextual meaning** in sentence completion
- **Synonym/antonym** via polarity (+/−) not rote lists

② READING COMPREHENSION

- **Read question stem first** for NAT/MCQ focus
- **Identify** main idea vs detail vs inference
- **Eliminate** extreme options
- **GATE RC** passages are short — don't overthink

③ NARRATIVE SEQUENCING / PARA JUMBLES

- **Opening sentence** = topic introduction
- **Pronouns/connectors** (this, such, however) bind order
- **Chronology** & cause → effect



If two options remain, pick the one supported by a line in the passage — not outside knowledge.

★ Context > memorised word lists.

04 — QUANT: %, RATIO & PROPORTION

Most GATE quant is arithmetic dressed as words — master % and ratio.

PERCENTAGE

- $x\%$ of $N = \frac{x}{100} \times N$
- % change $= \frac{\text{new} - \text{old}}{\text{old}} \times 100$
- Successive: $a\%$ then $b\%$
  $\rightarrow$ net $= a + b + \frac{ab}{100}$
- If A is $x\%$ more than B, B is $\frac{x}{100+x} \times 100\%$ less than A

RATIO

- $a : b = \frac{a}{b}$

Divide N in $a : b \rightarrow \frac{aN}{a+b}, \frac{bN}{a+b}$

- If $a : b = 2 : 3$ and $b : c = 4 : 5$
  $\rightarrow a : b : c = 8 : 12 : 15$

PROPORTION

- $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$
- Direct variation: $y \propto x$
  $\rightarrow y = kx$ (k is constant)
- Inverse variation: $y \propto \frac{1}{x}$
  $\rightarrow y = \frac{k}{x}$ (k is constant)

MINI WORKED EXAMPLES

1 35% of $240 = \frac{35}{100} \times 240 = 84$

2 A price increases from ₹800 to ₹920.
% increase $= \frac{920 - 800}{800} \times 100 = 15\%$

3 10% increase followed by 20% decrease.
Net change $= 10 - 20 + \frac{10 \times -20}{100} = -12\%$
Final value of 100 $= 100 \times (1 - 0.12) = 88$

4 Divide 450 in the ratio 2 : 3.
Parts $= \frac{2 \times 450}{5}, \frac{3 \times 450}{5} = 180, 270$

5 If $a : b = 3 : 4$ and $b : c = 5 : 6$, then $a : b : c = 15 : 20 : 24$

RED TRAPS — AVOID THESE!

1) Confusing "% of" with "% more"
- 20% of 100 = 20
- 20% more than 100 = 120 (not 20)

2) Wrong base in % change
- % change $= \frac{\text{new} - \text{old}}{\text{old}} \times 100$
- Not $\frac{\text{new} - \text{old}}{\text{new}} \times 100$

★ *Fix the base. Chain ratios carefully.* ★

05 — QUANT: AVERAGES, MIXTURES, WORK, SPEED

A • AVERAGE

- Average (mean) of n numbers:

$$\text{Avg} = \frac{\text{sum of values}}{n}$$

- Combined average (weighted):

$$\text{Combined Avg} = \frac{n_1A_1 + n_2A_2 + \cdots + n_kA_k}{n_1 + n_2 + \cdots + n_k}$$

- If average of n numbers is A, and one value x is replaced by y, then new average =

$$A + \frac{y - x}{n}$$

B • MIXTURE (ALLIGATION)



- By alligation, the ratio of quantities of cheaper : dearer = $(d - m) : (m - c)$
- Useful for milk–water mixtures, profit mixtures, concentrations, etc.

C • TIME & WORK

- If a work is finished in D days alone, work rate = $\frac{1}{D}$ work per day
- If A and B can do a work in A days and B days respectively, their work rate together =

$$\frac{1}{A} + \frac{1}{B} \text{ work per day}$$

- Pipes & Cisterns:
  Inlet = positive, Outlet = negative
  Rate = (sum of inlet rates) − (sum of outlet rates)

D • SPEED, DISTANCE, TIME

- Basic relation: $v = \frac{d}{t}$
- Average speed ≠ arithmetic mean
- For same distance covered at speeds u and v, average speed = $\frac{2uv}{u + v}$
- In a stream/river with speed of current = b, boat speed in still water = u
  - → Downstream speed = $b + u$
  - → Upstream speed = $b - u$

MINI TIP

Keep units consistent!
Convert when needed: 1 hour = 60 minutes, 1 day = 24 hours.

06 — QUANT: ALGEBRA & NUMBERS

Linear equations, identities, and number properties — keep it GATE-light, not JEE-deep.

IDENTITIES:

- $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

  $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

  $(a \pm b)^3$ etc briefly
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

EQUATIONS:

- Linear (two variables): $ax + by + c = 0$
  - Solve for one variable.
  - Age problems: let present age = x → express others → form equation.
- Quadratic: $ax^2 + bx + c = 0 \quad (a \neq 0)$
  - Sum of roots $(\alpha + \beta) = -\frac{b}{a}$
  - Product of roots $(\alpha\beta) = \frac{c}{a}$
  - Discriminant: $\Delta = b^2 - 4ac$

    $\Delta > 0$: real & distinct

    $\Delta = 0$: real & equal

    $\Delta < 0$: complex

INDICES & LOGS:

- $a^m \cdot a^n = a^{m+n}$
- $(a^m)^n = a^{mn}$
- $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$
- $\log(ab) = \log a + \log b$
- $\log\left(\frac{a}{b}\right) = \log a - \log b$
- $\log a^n = n \log a$
- Change of base:

  $\log_b a = \frac{\log_c a}{\log_c b} \quad (b>0, b\neq 1, c>0, c\neq 1)$

NUMBERS:

- Divisibility rules:

  2 : last digit even

  3 : sum of digits $\equiv 0 \pmod 3$

  4 : last two digits divisible by 4

  5 : last digit 0 or 5

  8 : last three digits divisible by 8

  9 : sum of digits $\equiv 0 \pmod 9$

  11 : (sum of digits in odd places) − (sum in even places) $\equiv 0 \pmod{11}$
- HCF × LCM = product of two numbers
- Remainders: If $N = dq + r$, then $r = N \bmod d$, $0 \leq r < d$

  (modular thinking — light level)
- Even / Odd (parity) tricks:

  even ± even = even, even ± odd = odd,

  odd ± odd = even, odd × odd = odd,

  even × any = even.

☆ — Setup equation cleanly; check extraneous roots. — ☆

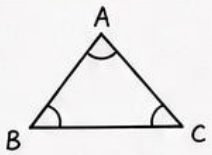
07 — QUANT: GEOMETRY & MENSURATION

★ Memorise a short formula sheet — GATE rarely needs heavy proofs.

1) PLANE GEOMETRY

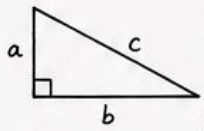
- Triangle angle sum

$$\angle A + \angle B + \angle C = 180°$$



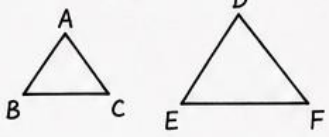
- Pythagoras Theorem

$$a^2 + b^2 = c^2$$



- Similar Triangles

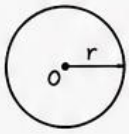
$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$



- Circle (basics)

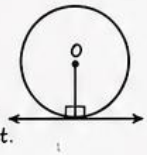
$C = 2\pi r$

$A = \pi r^2$



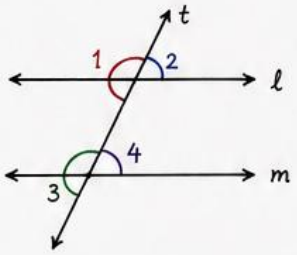
– Equal chords are equidistant from centre.

– Tangent ⊥ radius at point of contact.

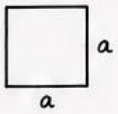


- Lines & Angles (Parallel lines cut by a transversal)

Alternate angles are equal.
Corresponding angles are equal.
Interior angles on same side are supplementary (sum = 180°).



2) MENSURATION (2D)



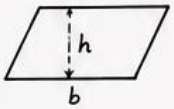
Square:

$$A = a^2$$



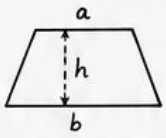
Rectangle:

$$A = l\,b$$



Parallelogram:

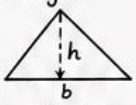
$$A = b\,h$$



Trapezium:

$$A = \frac{1}{2}(a+b)\,h$$

- Triangle:



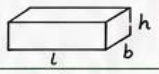
$$A = \frac{1}{2}\,b\,h$$

- Heron's Formula (for sides a, b, c)

$$s = \frac{a+b+c}{2},$$

$$\text{Area} = \sqrt{s\,(s-a)\,(s-b)\,(s-c)}$$

3) MENSURATION (3D)

Shape	Volume (V)	Total Surface Area (TSA)
Cuboid (ℓ, b, h) 	$V = l\,b\,h$	$TSA = 2(lb + bh + hl)$
Cube (a) 	$V = a^3$	$TSA = 6\,a^2$
Cylinder (r, h) 	$V = \pi r^2 h$	$TSA = 2\pi r\,(r + h)$
Cone (r, h) 	$V = \frac{1}{3}\pi r^2 h$	$TSA = \pi r\,(r + l)$ where $l = \sqrt{r^2 + h^2}$
Sphere (r) 	$V = \frac{4}{3}\pi r^3$	$A = 4\pi r^2$ (Surface Area)

Key Reminders:
Cylinder $V = \pi r^2 h$ Cone $V = \frac{1}{3}\pi r^2 h$ Sphere $V = \frac{4}{3}\pi r^3$, $A = 4\pi r^2$

4) COORDINATE GEOMETRY (Light)

- Distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Section Formula

$P(x_1, y_1)$, $Q(x_2, y_2)$, Point R divides PQ internally in ratio $m : n$ $(PR : RQ = m : n)$

$$R\left(\frac{mx_2 + nx_1}{m+n},\ \frac{my_2 + ny_1}{m+n}\right)$$



- Slope of line through P and Q

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (x_2 \neq x_1)$$

★ UNIT CONSISTENCY IS CRUCIAL:
For area → cm^2 vs m^2 | For volume → cm^3 vs m^3

Always convert to same unit before calculation.

★ Formula + diagram beat algebra thrashing.

08 — QUANT: P&C AND PROBABILITY

Count carefully — distinguish arrangements from selections.

PERMUTATION (P)

- $nP_r = \frac{n!}{(n-r)!}$ — order matters
- $n!$ arrangements of n distinct objects.
- Circular arrangements:
  $(n-1)!$ (fix one as reference)
  → Necklace: if reflections same, count ÷ 2 (when $n > 2$).
- Repetition allowed: n^r

COMBINATION (C)

- $nC_r = \frac{n!}{r!(n-r)!}$ — order irrelevant
- $nC_r = nC_{n-r}$
- $nC_0 = 1$ (and $nC_n = 1$)

- Pascal's identity:
  $nC_r + nC_{r-1} = (n+1)C_r$

PROBABILITY

- $P = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ (equally likely)
- $0 \le P \le 1$; $P(\emptyset) = 0$, $P(S) = 1$
- Complement: $P(A) + P(A') = 1$
- Independent events: $P(A \cap B) = P(A)P(B)$
- Mutually exclusive (disjoint):
  $P(A \cup B) = P(A) + P(B)$
- General case:
  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

MINI EXAMPLES

1 Coin (fair): $P(\text{Head}) = \frac{1}{2}$
$P(\text{2 Heads in 2 tosses}) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

---

2 Dice (fair): $P(3 \text{ or } 5) = \frac{2}{6} = \frac{1}{3}$
$P(\text{sum} = 7 \text{ with two dice}) = \frac{6}{36} = \frac{1}{6}$

---

3 Cards: From a deck of 52 cards
$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$
$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$

---

4 From 5 people, choose 2 (team):
$nC_r = 5C_2 = \frac{5!}{2!3!} = 10$
If order matters: $5P_2 = \frac{5!}{(5-2)!} = 20$

RED TRAPS — AVOID THESE!

1 Ordered vs Unordered
- Permutation (P): order matters
- Combination (C): order doesn't

---

2 With or Without Replacement
- With replacement: same item can be chosen again.
- Without replacement: once chosen, it cannot be chosen again.

---

3 Independence Trap
- Don't assume independence.
- Check if selections affect the outcome.

★ *Order? Replacement? Independence? Ask before computing.* ★

09 — QUANT: DATA INTERPRETATION

★ DI is reading graphs under time pressure — extract only what the question asks.

FORMATS

 table

 bar

 pie

 line

 mixed

METHOD

1. Read units & totals carefully
2. Note what is asked (% vs absolute)
3. Approximate when options are spaced
4. Cross-check pie slices sum to 100%

QUICK CALC

- $\% = \left(\frac{\text{part}}{\text{whole}}\right) \times 100$
- Growth rate year-on-year $= \left(\frac{\text{New} - \text{Old}}{\text{Old}}\right) \times 100$
- Ratio of two bars = A : B (same unit)
- Average of series = $\frac{\text{sum of values}}{\text{number of values}}$

TRAPS

- ✗ Wrong base year
- ✗ Ignoring "approximately"
- ✗ Mixing % points vs %
- ✗ Misread legends



TIP

For 2-mark DI, write one intermediate on rough sheet — prevents silly slips.



Units, base, legend — then compute.

10 — ANALYTICAL: LOGIC & DEDUCTION

Translate English to logic. Don't argue with real-world opinions.

① STATEMENTS: (Venn diagrams)

- All A are B



- Some A are B



- No A are B



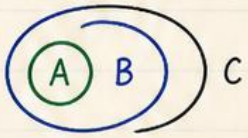
Conclusions:

✓ follow

✗ do not follow

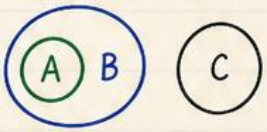
② SYLLOGISM: mini Venn sketches for common patterns

All A are B;
All B are C



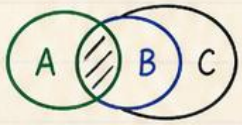
→ All A are C

All A are B;
No B are C



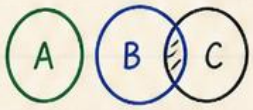
→ No conclusion

Some A are B;
All B are C



→ Some A are C

No A are B;
Some B are C



→ No conclusion

③ ASSUMPTION / INFERENCE / STRENGTHEN-WEAKEN

light for statement-based Qs

Assumption
- Hidden premise
- Needed for argument to hold

Inference
- Must follow from given
- No extra info

Strengthen
- Make argument stronger
- Increase certainty

Weaken
- Make argument weaker
- Reduce certainty

④ ANALOGY & MATCHING

Word analogy A : B :: C : ?
- Type of relation?

e.g., King : Queen :: Man : Woman

Number analogy patterns
- + / − / × / ÷
- Squares / Cubes
- Prime / Factor / Digit / Position

Odd-one-out
- Find the one that doesn't fit
- Check category, property, pattern

⑤ TRUTH TABLES light: AND, OR, NOT, implications for rare GATE logic Qs

P	Q	NOT P	P AND Q	P OR Q	P → Q	Q → P
T	T	F	T	T	T	T
T	F	F	F	T	F	T
F	T	T	F	T	T	F
F	F	T	F	F	T	T

NOT P : true when P is false
P AND Q : true only when both true
P OR Q : true when at least one true
P → Q : false only when P = T, Q = F

⚠ Some ≠ All. Possibility ≠ Certainty.

★ Draw Venns. Stick to given premises.

11 – ANALYTICAL: SERIES, CODING, PUZZLES

1 NUMBER SERIES:

- ★ ± arithmetic
- ★ × geometric
- ★ squares / cubes
- ★ alternate series
- ★ digit operations

Method: check difference → ratio → squares

2 LETTER SERIES:

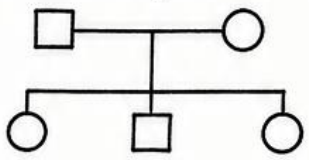
- Position in alphabet
  A = 1 ... Z = 26
- reverse Z = 1 sometimes
- +1 / +2 patterns; skip letters

3 CODING-DECODING:

- Letter shift cipher (e.g., A → D = +3)
- Symbol codes (e.g., @ = A, $ = S)
- Opposite letters
  A ↔ Z, B ↔ Y ...

4 BLOOD RELATIONS:

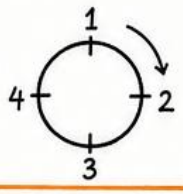
- Draw family tree



- don't assume gender from names alone if not stated
- Key: generation levels

5 SEATING / LINEAR & CIRCULAR:

- Fix one position (e.g., leftmost or any one person)
- Left / right from person's POV (facing direction is important)
- Keep relative positions fixed
- In circular, fix one person and face (usually) clockwise
- Take order always clockwise unless stated otherwise



If stuck > 90 seconds, mark and move — GA time is gold.

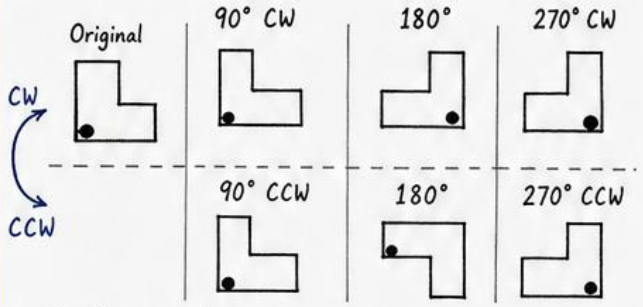
★ Pattern first, then options.

12 — SPATIAL APTITUDE

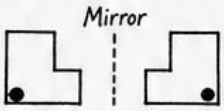
☆ Added to GATE syllabus — practise folding, rotation, and 3D views.

1) TRANSFORMATIONS

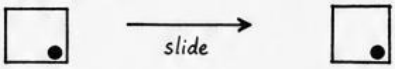
- Rotation 90° / 180° / 270° CW / CCW



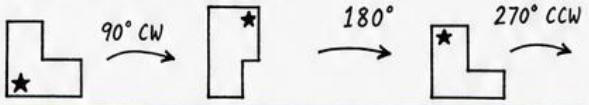
- Reflection / mirror



- Translation



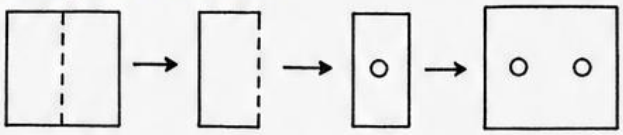
- Track a marked corner



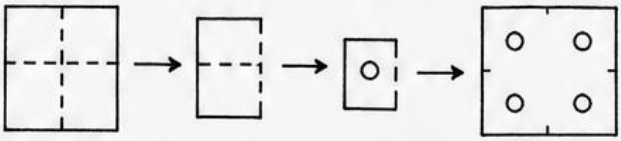
2) PAPER FOLDING & PUNCHES

- Fold → punch → unfold; punches multiply by fold lines.

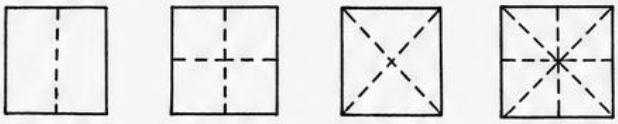
Example 1: fold once



Example 2: fold twice (perpendicular)



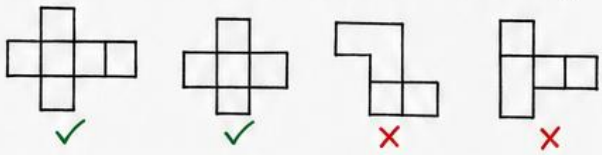
- Sketch fold axes



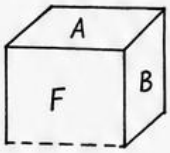
1 fold | 2 folds (⊥) | 2 folds (diagonal) | multiple folds

3) 2D → 3D

- Nets of cube / cuboid; which net folds validly.



- Opposite faces on a cube never share edge in net rules.

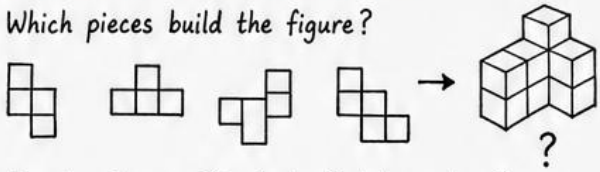


Opposite pairs:
A ↔ D (top ↔ bottom)
B ↔ F (front ↔ back)
C ↔ E (left ↔ right)

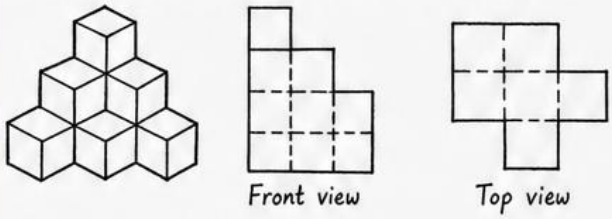
Use this rule to reject impossible nets.

4) ASSEMBLY / GROUPING

- Which pieces build the figure?

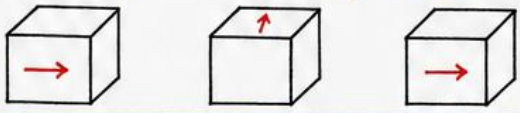


- Count cubes in 3D stack (hidden cubes!)



Total cubes = 14 (don't forget the hidden ones!)

💡 Practice tip: draw on rough sheet; mark arrows on faces.



Keep a reference corner (★) and track it consistently.

⚠ Hidden cubes in stacks are the classic trap.

☆ Mark a reference corner. Count what you cannot see.

13 — FORMULA CHEAT SHEET

% SUCCESSIVE (%)

- Increase:
  $\text{New} = \text{Old}\left(1+\frac{r_1}{100}\right)\left(1+\frac{r_2}{100}\right)\cdots$
- Decrease:
  $\text{New} = \text{Old}\left(1-\frac{r_1}{100}\right)\left(1-\frac{r_2}{100}\right)\cdots$
- Net % change:
  $\left[\frac{(\text{Final}-\text{Initial})}{\text{Initial}}\right]\times 100$

% MORE / LESS (%)

- $x\%$ more than $y = y\left(1+\frac{x}{100}\right)$
- $x\%$ less than $y = y\left(1-\frac{x}{100}\right)$

- More ↔ Less reverse:
  If A is $x\%$ more than B, then B is $\frac{x}{100+x}\times 100\%$ less than A.

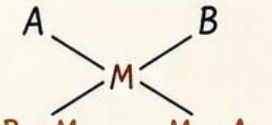
RATIO COMBINE

- If $a:b = m:n$ and $b:c = p:q$, then $a:b:c = m:n:q\times p$

- If $a:b = x:y$, $b:c = y:z$ then $a:b:c = x:y:z$

ALLIGATION

For mixture of costs (or strengths) A and B to obtain mean M



Ratio of $A:B = (B-M):(M-A)$

AVERAGE SPEED

For equal distances with speeds u & v

$$v_{avg} = \frac{2uv}{u+v}$$

For unequal distances (d_1, d_2):

$$v_{avg} = \frac{d_1+d_2}{\frac{d_1}{u}+\frac{d_2}{v}}$$

WORK & TIME

- Work = Rate × Time
- If A alone does in x days, rate $= \frac{1}{x}$
- If A, B together in x days: rate $= \frac{1}{x}$
- If A in x days, B in y days together:
  Time $= \frac{xy}{x+y}$ days

$(a+b)^2$

$$(a+b)^2 = a^2+2ab+b^2$$

a^2-b^2

$$a^2-b^2 = (a+b)(a-b)$$

LOGS ($a>0, a\neq 1$)

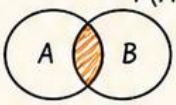
- $\log_a(MN) = \log_a M + \log_a N$
- $\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$
- $\log_a(M^k) = k\log_a M$
- $\log_a 1 = 0$
- $\log_a a = 1$
- Change of base: $\log_a N = \frac{\log_b N}{\log_b a}$

${}^nP_r$ and ${}^nC_r$

- ${}^nP_r = \frac{n!}{(n-r)!}$
- ${}^nC_r = \frac{n!}{r!\,(n-r)}$
- ${}^nC_r = {}^nC_{n-r}$
- ${}^nP_r = {}^nC_r \times r!$
- $(n+r-1)C_r = {}^nH_r$ (repetition)

P(A ∪ B)

- $P(A\cup B) = P(A)+P(B)-P(A\cap B)$



- If A, B are mutually exclusive $(A\cap B=\phi)$:
  $P(A\cup B) = P(A)+P(B)$

INDEPENDENT EVENTS

- Independent if $P(A\cap B) = P(A)\,P(B)$

- For independent A, B:
  $P(A/B) = P(A)$
  $P(B/A) = P(B)$

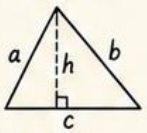
HCF × LCM

For any two positive integers a, b:

$$HCF(a,b)\times LCM(a,b) = a\times b$$

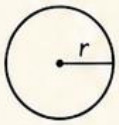
GEOMETRY (Key Formulae)

TRIANGLE



- Area $= \frac{1}{2}bh$
- Perimeter $= a+b+c$
- Heron's formula:
  $s = \frac{a+b+c}{2}$
  $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

CIRCLE



- Circumference $= 2\pi r$
- Area $= \pi r^2$
- Arc length $= \frac{\theta}{360}\times 2\pi r$
- Sector area $= \frac{\theta}{360}\times \pi r^2$

CYLINDER



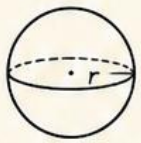
- Curved surface area $= 2\pi rh$
- Total surface area $= 2\pi r(h+r)$
- Volume $= \pi r^2 h$

CONE



- $l^2 = r^2+h^2$
- CSA $= \pi rl$
- TSA $= \pi r(l+r)$
- Volume $= \frac{1}{3}\pi r^2 h$

SPHERE



- Surface area $= 4\pi r^2$
- Volume $= \frac{4}{3}\pi r^3$

DISTANCE FORMULA

For points $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

★ Revise this page every week until the exam. ★

14 — STRATEGY & TIME MANAGEMENT

GA is 15 marks — aim to finish in 20–25 minutes with high accuracy.

ATTEMPT ORDER

1. Easy quant NAT/MCQ (%, ratio, averages)
2. Grammar / vocab one-liners
3. Short analytical
4. DI / spatial
5. RC last among GA if lengthy

NEGATIVE MARKING DISCIPLINE

MCQ: attempt only if you can eliminate to ≤2 options confidently

MSQ: select only sure options — partial wrong can zero the question

NAT: no negative — attempt all NATs with care

PREP PLAN (GATE 2027)

- Daily: 20–30 min GA until exam
- Weekly: 1 full GA section timed
- Monthly: analyse error log (silly vs concept)

TARGET

- Safe: 10+/15
- Good: 12+/15
- Excellent: 14–15

Technical paper mood: don't bleed 30 minutes into one RC.

Accuracy compounding beats heroic guesses.

15 — PRACTICE PATTERNS & FINAL RECAP

HIGH-FREQUENCY PATTERNS:

1	% successive & population growth
2	Ratio salaries / mixtures alligation
3	Time-work pipes
4	Speeds avg & upstream
5	Grammar SVA + preposition
6	Vocab in context
7	Syllogism Venn
8	Number/letter series
9	Pie/bar DI
10	Cube nets / paper folds

RESOURCES HABIT:

- Previous GATE GA papers (last 10 years)
- Official syllabus checklist tick-off
- Error notebook

MINDSET:

- GA is not “too easy to prep” — toppers bank it
- Sleep well before exam; read each stem twice on NAT

BANK THE 15. PROTECT THE RANK.